

Introduction

- In the wake of the COVID-19 pandemic, the number of people worldwide living below the poverty line increased by 90 million in 2020¹.
- Financial difficulties can perpetrate further issues, such as housing insecurity, which can be divided into household changes and housing quality.
- Our stress response system is primarily mediated by the hypothalamic-pituitary-adrenal (HPA) axis.
- With a decrease in household quality of life, comes physical and mental problems as well².



Figure 1. HPA Axis Pathway. Various physiological changes focus the body on homeostasis and survival only³. Overstimulation can result in neuroendocrine issues and mental health disorders⁴.

Objective: Investigate whether children exposed to housing insecurity indices experience dysregulation of their HPA axis with regard to cortisol levels.

Methods

- MedLine, PsycInfo, Web of Science, and PubMed databases were used in the search.

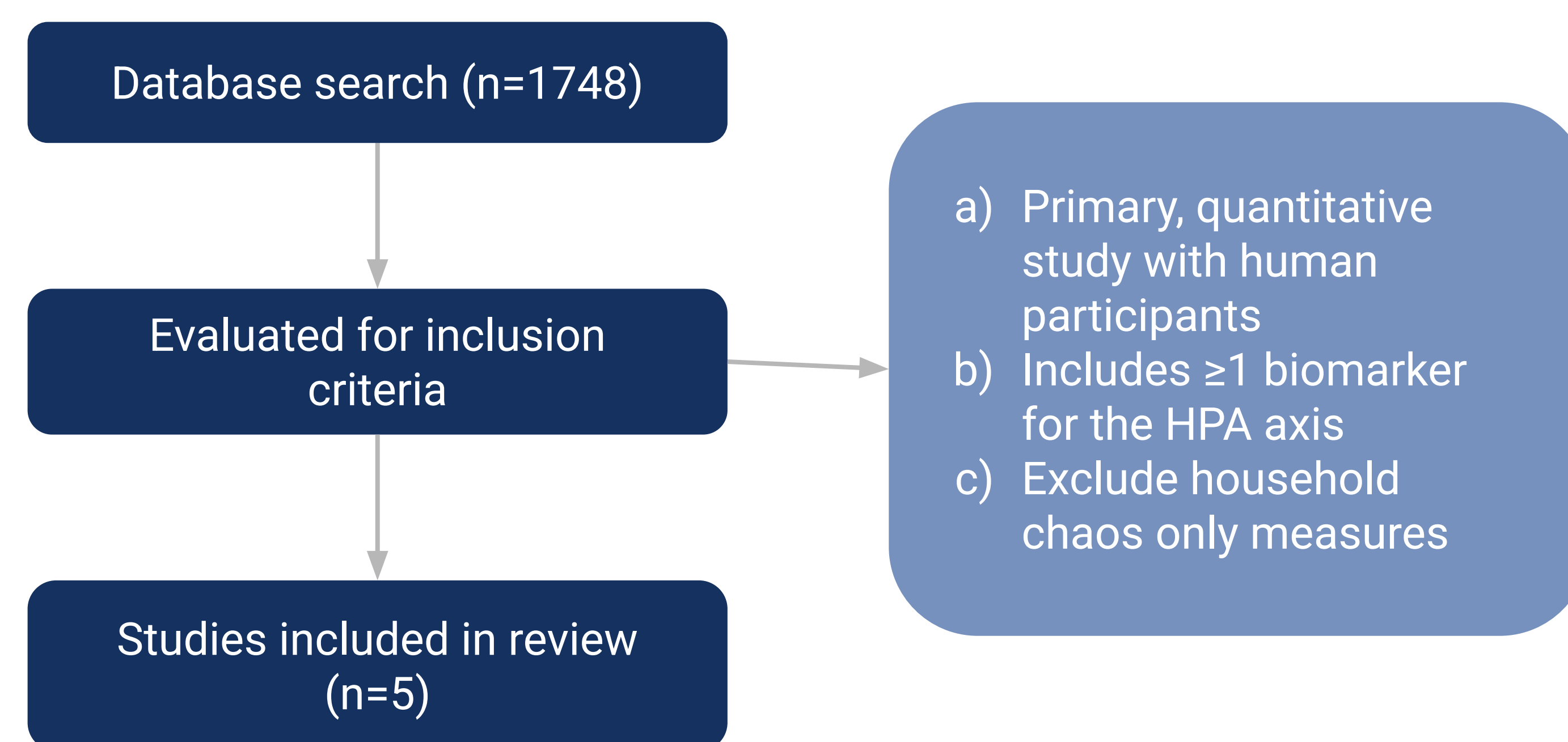


Figure 2. Search Procedure and Inclusion Criteria Flow Diagram. Search terms were developed in accordance with the PICOT method. Studies were obtained from PsycInfo (2) and PubMed (3).

Results

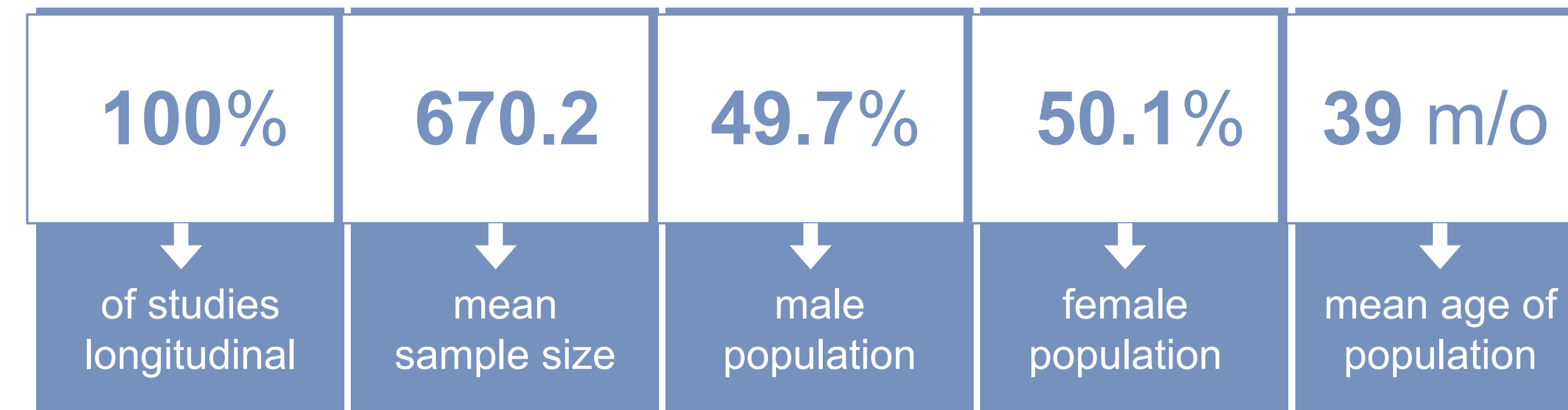


Figure 3. Demographics of Records Analyzed. Mean sample size (SD=569.91), mean percentage of males in the population (SD=1.35), mean percentage of females in the population (SD=1.35), and mean age in months (SD=14.17)

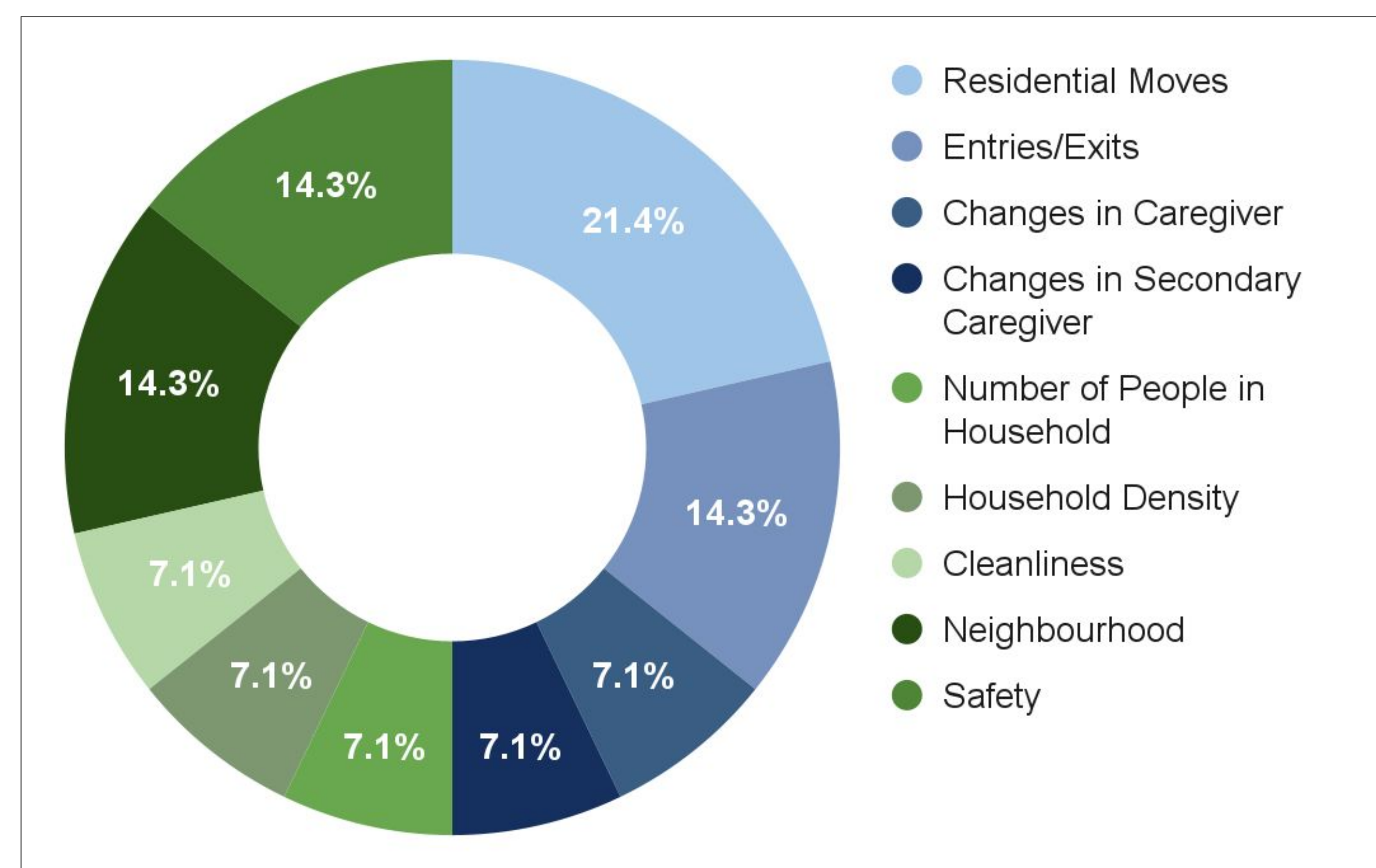


Figure 4. Housing Instability Index Measures Across Studies. Each study analyzed in this review utilized a variety of different measures for household instability.



Figure 5. Regression Coefficient Between Housing Insecurity Indices and Cortisol Levels. The relationship between household insecurity indices, divided into household changes and housing quality. This varied between positive and negative correlations ($p < 0.05$).

Discussion

- Evidence from the records examined display the effect of housing instability indices on child HPA axis functioning.
- Analysis of salivary cortisol as a biomarker of the HPA axis displays dysregulation of the stress response system for increased household changes and decreased housing quality.
- This presents itself in the form of both hyperarousal and hypoarousal or blunted levels of cortisol, which can have severe long-term effects due to desensitization or overstimulation of the HPA axis.
- This review provides unique insight into the intersection of social and biological aspects of socioeconomic status.

Limitations/Future Directions

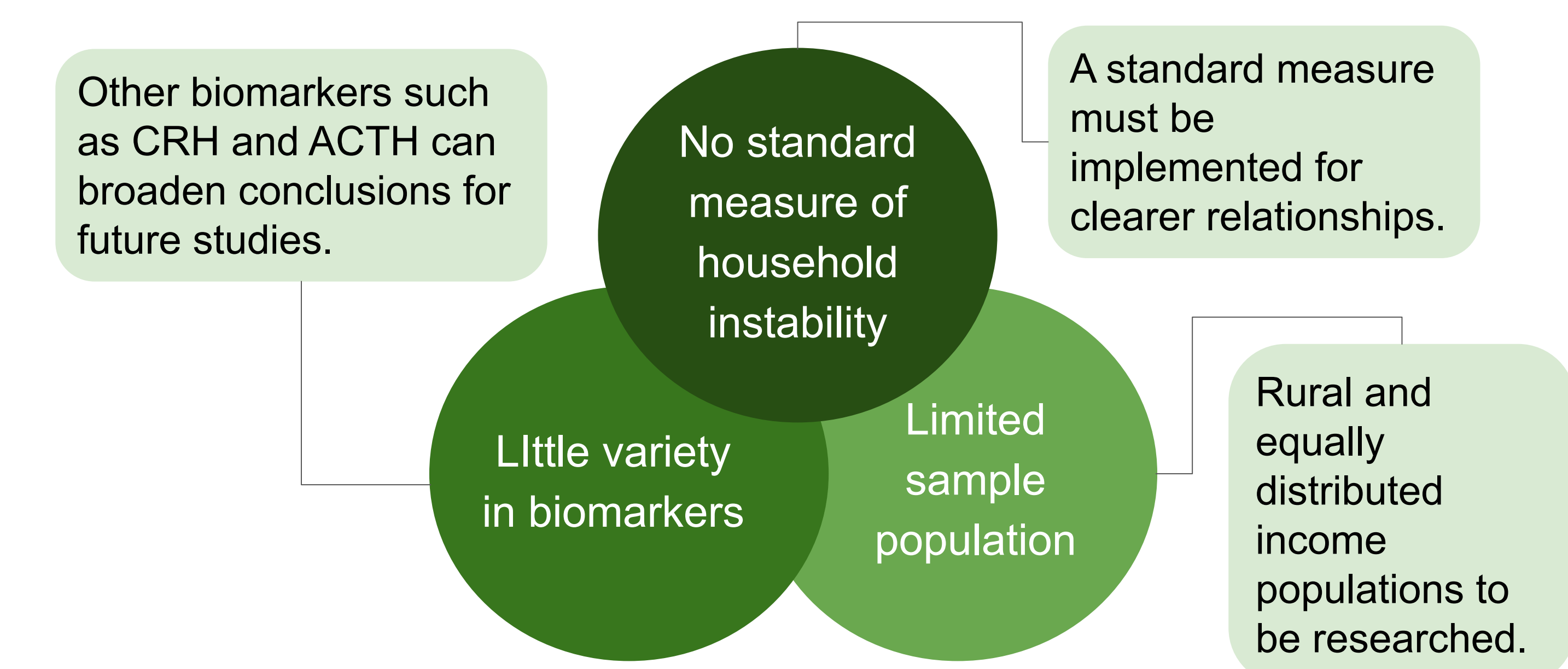


Figure 6. Future Directions Diagram. Limitations found between the studies analyzed and potential future studies that would be able to draw more applicable and strong conclusions about the relationships discussed.

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*studies analyzed for the purpose of the literature review